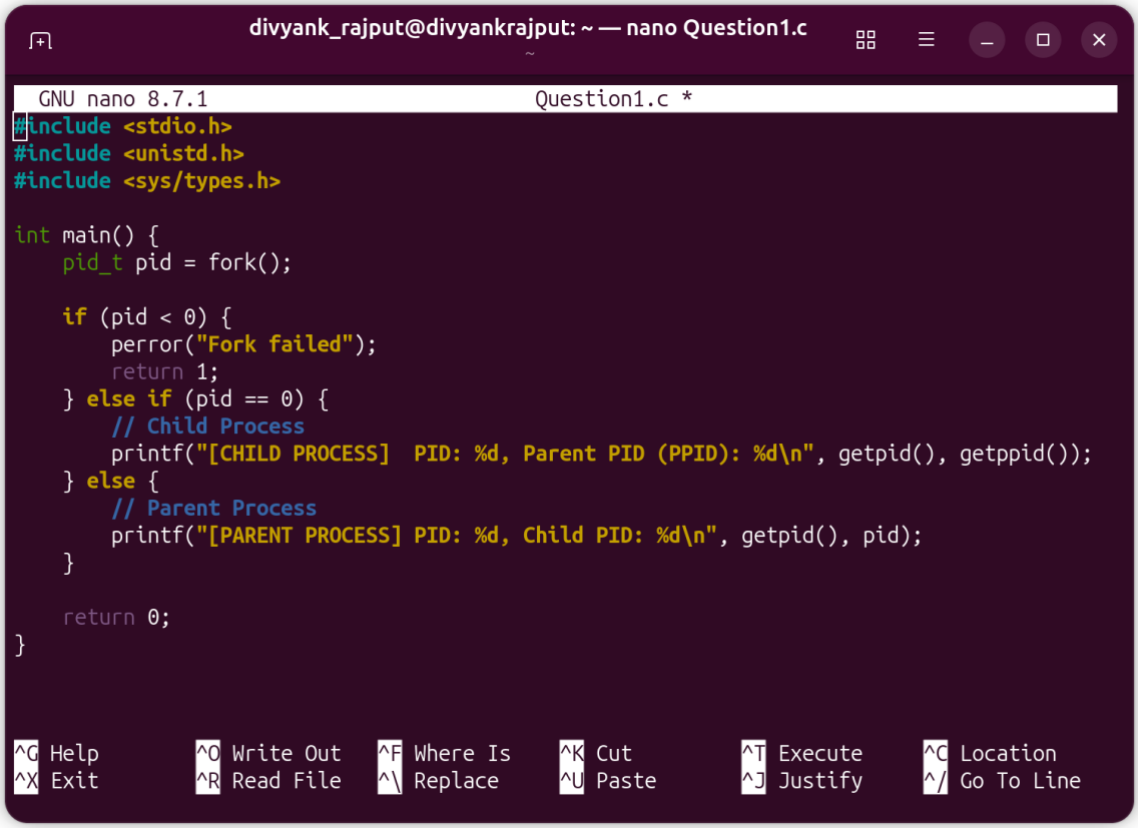


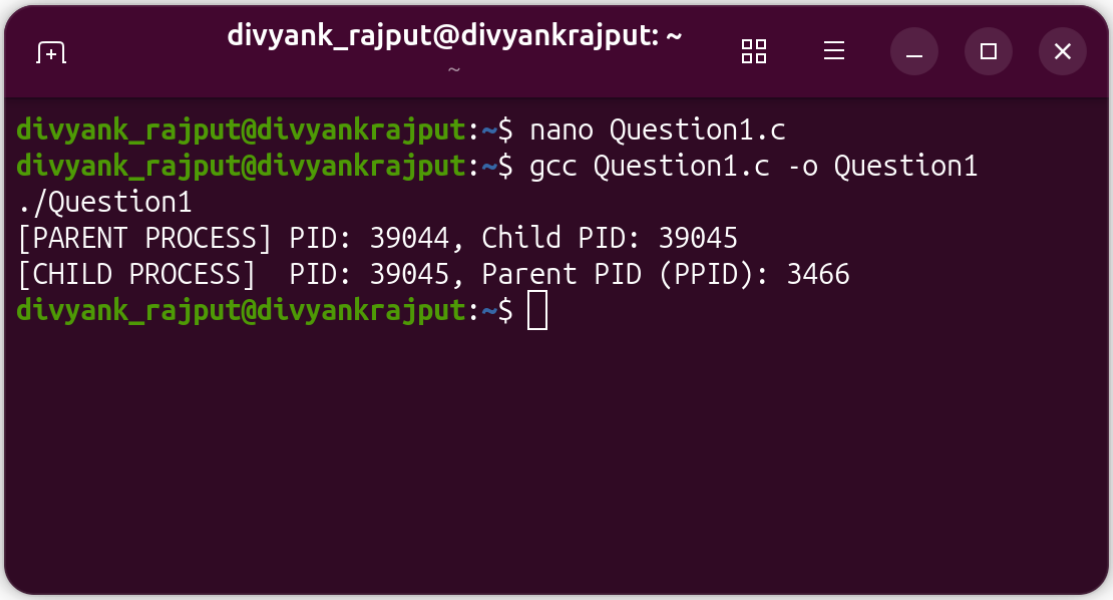
THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

Assignment 5:
Operating Systems
(MCA-103)

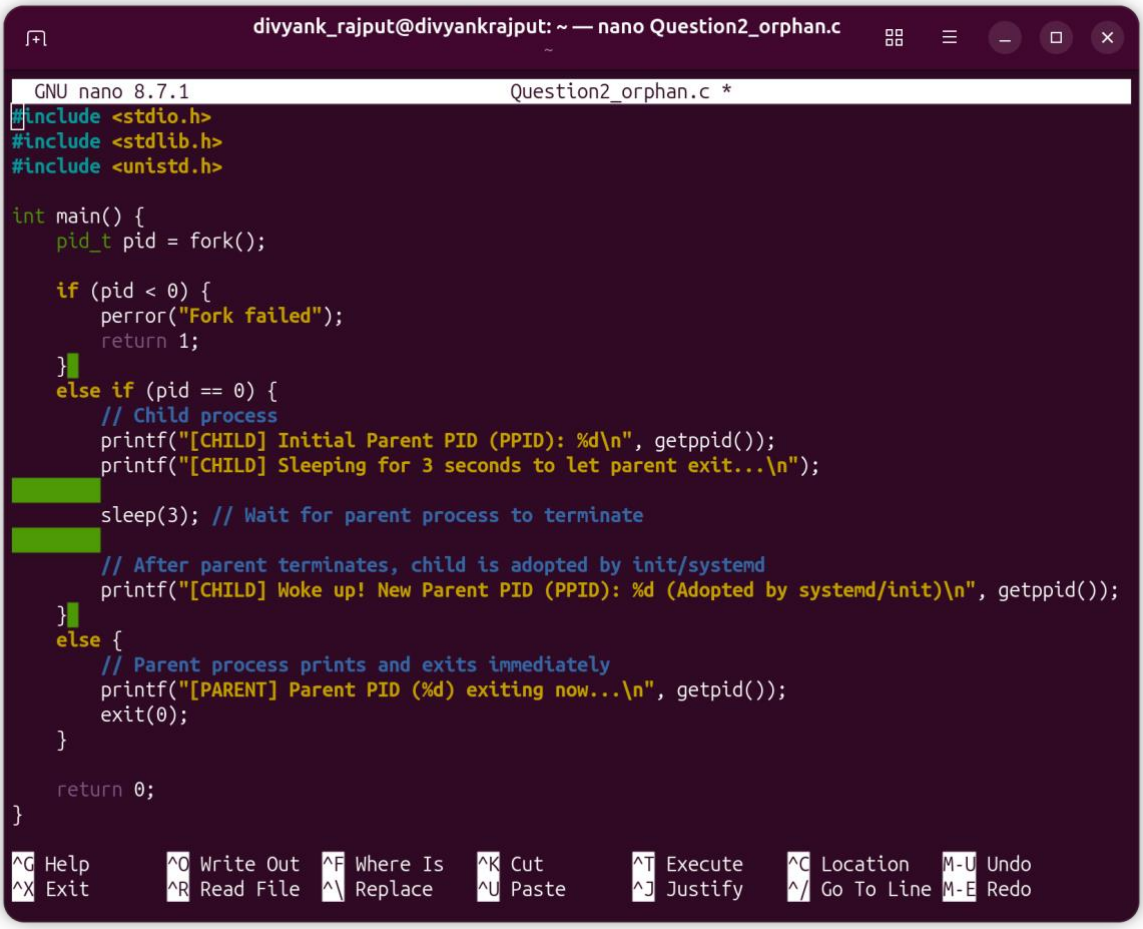
Divyank Rajput
2026010038
1MCA-B

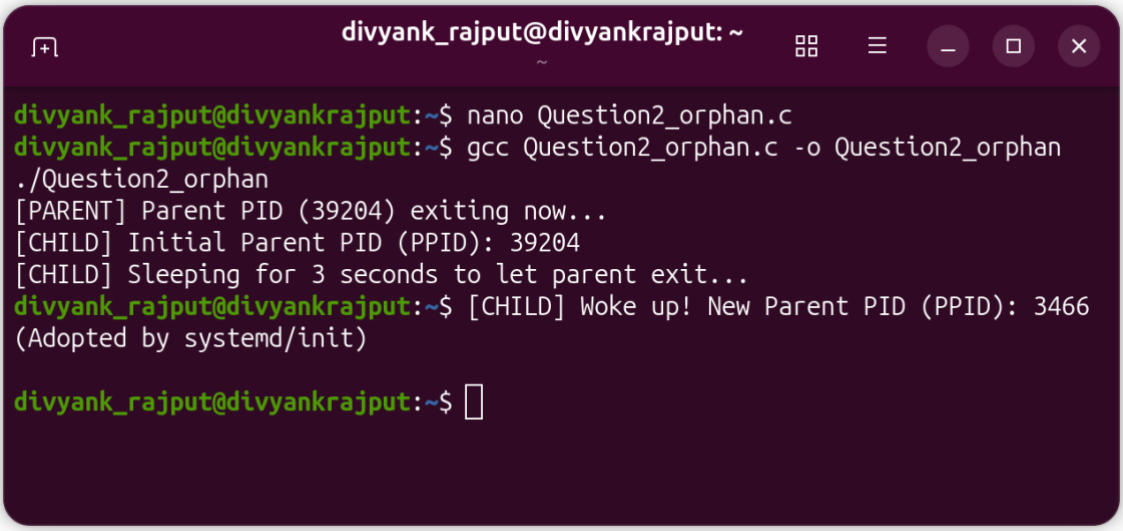
Q1. Write a program to:
Create a child process using fork().
Print the process IDs (PID, PPID) in both parent and child.



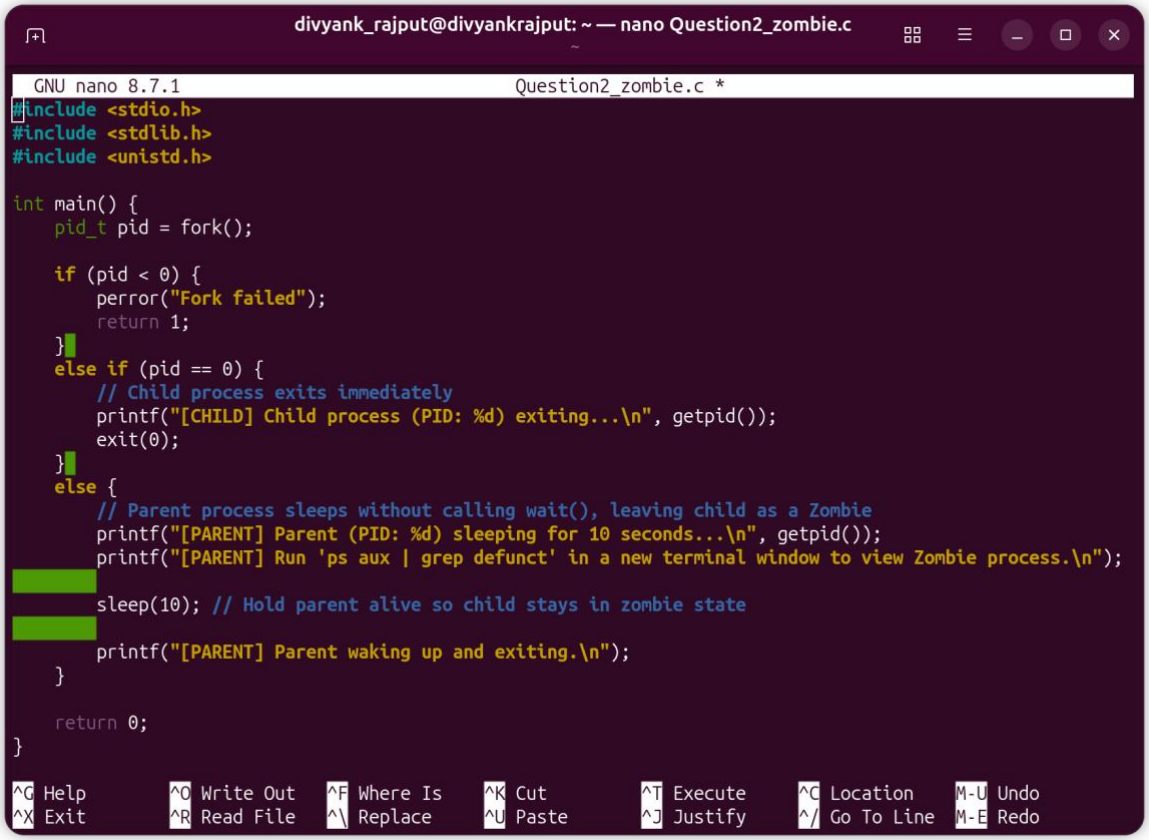


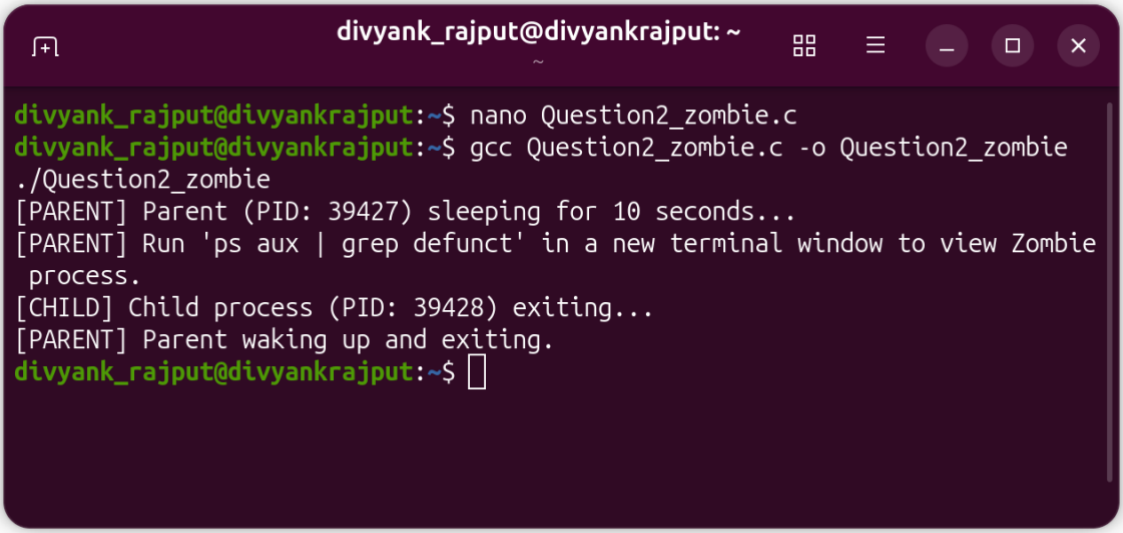
Write a program to:
Demonstrate creation of an orphan process using fork() and sleep().





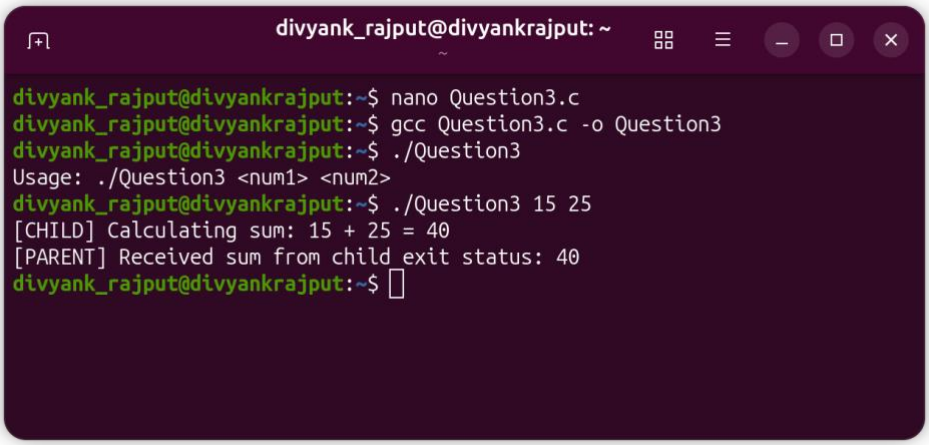
Demonstrate creation of a zombie process (child exits before parent calls wait ()).





Q3. Write a program that accepts two small numbers (< 50) as arguments and then sums the two in a child process. The sum should be returned by the child to the parent as its exit status and the parent should print the sum.





Q4. Predict the output of the following program.

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
int main() {
int x = 10;
pid_t pid;
pid = fork(); if (pid < 0) { perror("fork failed");
// create child process
// error in fork
return 1;
}
if (pid == 0) { printf("Child: Initially x = %d\n", x);
x = x + 5; printf("Child: Modified x = %d\n", x);
} else {
// Child process
// modify x in child
// Parent process
// wait for child to finish
wait(NULL); printf("Parent: x = %d\n", x);
}
return 0;
}
```

Ans:

Child: Initially x = 10
Child: Modified x = 15
Parent: x = 10

Q5: Demonstrate use of lseek() System call.

Function Prototype:

off_t lseek(int fd, off_t offset, int whence);

Parameters:

• fd → File descriptor (from open())
• offset → How far to move the pointer
• whence → Reference point for movement:
o SEEK_SET → Beginning of file
o SEEK_CUR → Current file position
o SEEK_END → End of file

Create a file 'seeking' and write some content e.g. 1234567890abcdefghijklmnopqrstuvwxyz and save it. Predict the output of following

code:

```
#include <sys/types.h>
#include <sys/stat.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdio.h>
int main() {
int f;
char buff[10];
f = open("seeking", O_RDWR); if (f < 0) {
perror("open");
return 1;
// open the file "seeking" in read+write mode
}
// read first 10 bytes and print
read(f, buff, 10);
write(1, buff, 10);
// move file pointer 10 bytes ahead from current
lseek(f, 10, SEEK_CUR);
// read next 10 bytes and print
read(f, buff, 10);
write(1, buff, 10);close(f);
return 0;
}
```

Ans:

1234567890klmnopqrst

Q6. Predict output of following codes as well for the seeking file mentioned in Q5.
Prog1.
Ans:
Pointer is at 10 position
abcdefghij

Prog2.
Ans:
pqrstuvwxy

Q7. Predict output of the following programs.

Ans:
11
10